Venkata Lakshmi Parimala Pasupuleti

AI Solution Architect · Agentic AI Platform · LangGraph · Model Evaluation · Azure AI · MLOps · Python · Responsible AI · Data Governance

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PROFESSIONAL SUMMARY

AI Solution Architect designing and building enterprise agentic AI platforms — the exact architecture EisnerAI requires. LeaseI is an 8-agent LangGraph compliance system (state management, logging, error handling, HITL escalation) solving the same problem TaxPrepBot and ComplianceRiskBot will solve at EisnerAmper. AgentProbe is a production model evaluation framework driving post-training optimization: task completion rate, hallucination rate, CI regression harness, LLM-as-judge, golden datasets. Integration across AI, Agents, Application, and Data layers: Azure (Databricks, AML/AI Foundry, ADF, ADLS, OpenAI Service, AI Search, Cognitive Services), AWS, PostgreSQL, Python, C#, TypeScript. Data governance: PHI/HIPAA/SOC 2/NIST/OWASP LLM Top 10/audit logging. M.S. Data Analytics Engineering, GMU (GPA 3.87). — github.com/venkatalakshmiparimala/AgentProbe

WORK EXPERIENCE

Blue Horse Digital
Sep 2026 – Present

AI/ML Engineer (Volunteer) · Remote

- Designing AI architecture and ML models; building agentic pipelines that integrate with backend APIs; applying responsible AI practices and data governance policies; MLOps and model lifecycle management across deployment and monitoring.

Magizhs Tech Solution
May 2025 – Aug 2025

AI/ML Engineer Intern · Remote

- Architected and led full AI agent platform: designed agentic framework (LangGraph) covering initiation, execution, state management, logging, and error handling; RAG layer over regulatory corpus; safety guardrails as hard enforcement logic; responsible AI practices enforced throughout. Owned the integration across AI, Agents, and Application layers end-to-end.

George Mason University
Aug 2024 – Aug 2025

Research Associate + Teaching Assistant

- Built multi-source ML prediction pipelines (Python, R) across 190+ countries ($R^2 = 0.977$); HuggingFace Transformers fine-tuning + PyTorch MLOps (model training, evaluation, deployment lifecycle); Isolation Forest anomaly detection; statistical modeling. Workshops to 600+ students across Python and SQL.

Entrata (PropTech / Real Estate SaaS)
Sep 2018 – Aug 2024

Senior Software Engineer · Remote

- **Enterprise platform architecture and data governance** (Python, Java, C#, TypeScript, PostgreSQL, RabbitMQ, AWS, Kubernetes) — owned 20+ production modules at 99.5% uptime for 100K+ MAU. Designed integration across application, data, and service layers; enforced data governance policies and compliance standards (PHI, HIPAA, SOC 2, zero security incidents across 7 years); Libsodium-encrypted audit pipelines; AWS (S3, EC2, RDS, Lambda, Step Functions); Docker + Kubernetes.
- **AI/ML data infrastructure and post-training pipelines** — Azure Databricks data engineering pipelines; Azure Data Factory (ADF) and Azure Data Lake Storage (ADLS) for enterprise-scale data management; 200+ PostgreSQL optimizations (30% performance gain); schema migrations across hundreds of thousands of records; OpenTelemetry + Prometheus + Grafana observability; CI/CD (GitHub Actions, Azure DevOps-compatible) with automated testing; Python/Bash automation eliminating 80% of manual workload.

PROJECTS

AgentProbe: Model Evaluation Framework & Post-Training Optimization
LLM-AS-JUDGE MLOPS

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions · OpenTelemetry | github.com/venkatalakshmiparimala/AgentProbe

- **Model evaluation framework driving post-training optimization**: task completion rate, hallucination rate, groundedness, latency, token efficiency; LLM-as-judge classifiers for automated quality scoring; **CI regression harness with golden datasets** ensuring model improvements don't degrade production behavior; multi-provider LLM routing (OpenAI/Anthropic/Groq/Ollama) with cost/latency controls; offline + online evaluation loops; trace-level observability for auditable AI. Exactly "drive model evaluation frameworks and post-training optimization for custom LLMs and AI agents."

Leasel: Enterprise AI Agent Workforce - TaxBot / ComplianceBot Architecture
LANGGRAPH RESPONSIBLE AI

Python · LangGraph · FastAPI · Azure OpenAI Service-compatible · ChromaDB · MCP · LangSmith | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Production AI agent workforce: 8-agent LangGraph platform (Triage → Retrieval → Configuration → Historical → Reasoning → Validation → Escalation) with **initiation, execution, state management, logging, and error handling** — the exact agentic framework design EisnerAmper specifies. Responsible AI: regulated outcomes always route to human review (HITL); every answer cited to source; safety guardrails as code-enforced logic. Evaluated with AgentProbe: 93.6% precision, 87.9% F1 across 1,240 compliance test cases.

EDUCATION

M.S. Data Analytics Engineering George Mason University, Virginia — May 2026

GPA: 3.87

B.Tech Computer Science & Engineering JNTU Anantapur — 2017

SKILLS

Agentic AI & Evaluation: LangGraph, LangChain, LangSmith, MCP, AI Agent Workforce, Custom LLMs, Agentic Frameworks, State Management, Initiation & Execution, Error Handling, HITL Escalation, Safety Guardrails, Responsible AI, Data Management Layer, Model Evaluation Frameworks, Post-Training Optimization, LLM-as-Judge, Hallucination Detection, CI Regression Harnesses, Trace-Level Observability, Audit Logging, RAG Pipelines, Knowledge Mining, Vector Search, Semantic Search, ChromaDB, Qdrant, Pinecone, Weaviate, GPT Models, Model Fine-Tuning, Prompt Engineering, Prompt Injection Defense, OWASP LLM Top 10, OpenAI API, Anthropic API

Azure & Cloud AI: Azure AI Foundry, Azure Machine Learning (AML), Azure OpenAI Service (GPT fine-tuning), Azure AI Search (vector search + semantic search + knowledge mining), Azure Cognitive Services, Azure Document Intelligence, Azure Language Understanding, Azure Computer Vision, Azure Databricks, Azure Data Factory (ADF), Azure Data Lake Storage (ADLS), Azure Cosmos DB, Azure DevOps, MLOps, AWS (S3, EC2, RDS, Lambda, Bedrock, SageMaker), GCP, Vertex AI, Docker, Kubernetes (K8s), Terraform, CI/CD (GitHub Actions), OpenTelemetry, Prometheus, Grafana

Languages & Data: Python (8+ yrs), C#/.NET, TypeScript, JavaScript, Java (Spring Boot), SQL, Bash — PyTorch, TensorFlow, Scikit-learn, Transformers, HuggingFace, BERT, Computer Vision, Language Understanding — PostgreSQL, MySQL, Redis, Apache Spark, PySpark, FastAPI, REST APIs, Data Governance, PHI Compliance, HIPAA, SOC 2, NIST